RESEARCH REPORT · VITKI

Partially tested finding

Language and content order the runestones of Uppland in the same sequence as the stylistic chronology – rank correlation 0.54 without the ornament

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Timeline of the study: The seriation was first made 5 October 2026 · The material was frozen 29 October 2026 · The plan for the halves and Södermanland was locked and run 31 October 2026

ABSTRACT

What has not been tested. The seriation in Uppland was not pre-registered. Of the fixed series of sensitivity analyses only the halves are tested, under a plan locked before the run together with a replication in Södermanland; the carver and the parish as random effects, location as a spatial surface, the analysis without moved stones and the model check are not tested.

Background and aim. The runestones of Uppland are dated relatively by their ornament, following Anne-Sofie Gräslund's stylistic chronology. The question here is whether the stones fall in the same sequence when ordered by something else: the language, the formulas and the content.

Material. 818 runestones in Uppland with at least four of 65 traits, 501 of them with a style group; for the replication 270 stones in Södermanland, 151 with a style group. From the Scandinavian Runic-text Database (Rundata). The seriation is made on Uppland, since in the whole Swedish material its first axis mainly follows regional differences.

Method. Correspondence analysis of a table of stones and traits, without the style group; the stones' position on the first axis is compared with the style group's midpoint (Spearman's rank correlation). The halves and Södermanland with the same method, Holm-corrected.

Results. In Uppland rho = 0.54, 0.54 with latitude held constant. Half A 0.51, half B 0.54, both Holm-corrected p below 10^{-17}. Södermanland 0.32, Holm-corrected p = $6.3 \cdot 10^{-5}$, but 0.29 with latitude held constant. Within hundreds 0.51, within one and the same carver only 0.25.

Conclusion. Language and content carry the same temporal sequence as the ornament in Uppland. That is support for the stylistic chronology that does not rest on the ornament. Within one and the same

carver the association is weak, but each carver's stones span a short time, so the material cannot tell how much of the agreement runs through the carvers. It is not enough to date individual stones.

Keywords: *runestones, Viking Age, seriation, correspondence analysis, stylistic chronology, dating, Uppland, Södermanland*

HIGHLIGHTS

- What is written on the stones changed in the same order as the ornament.
- That gives the stylistic chronology support that does not rest on the ornaments themselves.
- Within one and the same carver the order is weak, but a carver's stones span a short time, so that says little. The order is not enough to date a single stone.

1. Introduction

The Viking-Age runestones of Uppland are dated relatively by their ornament, in Anne-Sofie Gräslund's style groups from RAK and Fp to Pr1–Pr5 (Gräslund 2006). The chronology rests on how the rune animal and the animal heads are shaped, not on the language or on what the text says. If language and content changed over the same time, the stones should fall in roughly the same sequence when ordered by them. If two kinds of observation give the same order, both probably follow time – unless both follow something third, such as the carver.

Single traits have earlier been set against the style groups, for instance how ʀ is written in Södermanland and Uppland style group by style group (Larsson 2002, tables 6–9). Seriation by correspondence analysis is a common method of relative dating in archaeology (Greenacre 2007). We have found no study that orders many traits at once, without the ornament, and tests the order in two halves of the material and in a second province, Södermanland, where the chronology was not built.

2. Pre-registration: what was fixed in advance and what is exploratory

The seriation in Uppland was first made on 5 October 2026 as part of the project's survey, without a locked plan, and its value was known when the rest of the study was planned. It is therefore not pre-registered. The same survey had shown that the seriation's first axis in the whole Swedish material follows regional differences (3), and it was therefore made on Uppland. The variants without crosses, without journeys abroad and without the traits that change over time, and the correlations within hundreds and carvers (5.3), were computed at the same time and are exploratory.

On 31 October 2026 a plan for two tests was locked before they were run (SHA-256 checksum `454-cacf9d8185a26…`): the seriation in each half of the stones of Uppland on its own, and the same seri-

ation on the stones of Södermanland, which had not been seriated on their own. The halves hold when Holm p is below 0.05 and rho at least 0.3 in both. Södermanland is *replicated* by the same limits; if the correlation with latitude held constant is below 0.3, the association is said to be possibly geographical. A trial run reproduced Uppland's known value exactly before the tests were run.

3. Material

- **Source.** The Scandinavian Runic-text Database, Department of Scandinavian Languages, Uppsala University: Rundata (2014 edition, updated 2018), retrieved 2026-10-04; derived corpus computed 2026-10-29. Runestones only.
- **The traits.** Language traits (the spelling of certain sounds and words), five formulas (sponsor formula, monument, word order, carver's signature, prayer), cross, short-twig runes and content categories, as presence or absence per stone. Traits found on fewer than 15 stones are not counted, nor stones with fewer than four traits. The style group is not included.
- **Time** is the style group's midpoint after Gräslund (2006).
- **Why Uppland.** In the whole Swedish material (1,629 stones with at least four traits) the seriation's first axis follows mainly regional differences and the carvers, and time only weakly (rank correlation with the style-group midpoint 0.17); the second axis follows time better (0.59), but also province and carver. The seriation is therefore made on Uppland, where the stylistic chronology was built.
- **Uppland:** 818 stones and 65 traits, 501 stones with a style group. **Södermanland:** 270 stones and 44 traits, 151 with a style group.
- **The halves** are the project's fixed division parish by parish.

4. Method

4.1 The seriation. Correspondence analysis (the package ca; Nenadić & Greenacre 2007) of the table of stones and traits. The analysis places stones that have the same traits close together, and traits that tend to occur on the same stones close together; the first axis is the direction in the material that separates the stones most. The stones' position on the first axis is compared with the style group's midpoint by Spearman's rank correlation. The sign of the axis is arbitrary and is turned so that the correlation is positive.

4.2 The geography. Since language traits also differ between north and south, the partial correlation with latitude held constant is reported.

4.3 The halves and Södermanland. The same method, unchanged, on each half and on Södermanland, Holm-corrected over the three tests.

4.4 The variants (exploratory). The seriation without the trait cross, without the trait journey abroad, without the traits whose change over time is measured separately (dotted runes, the monophthong, the ós-rune) and without all of these; the variants drop the traits, not the stones. In addition, the correlation within hundreds and within carvers with at least five dated stones.

4.5 Deviations. None from the plan.

5. Results

5.1 Uppland and the halves. In Uppland rho = 0.54 (uncorrected p = $9.4 \cdot 10^{-40}$) and 0.54 with latitude held constant; latitude itself correlates only 0.09 with the axis. Half A: 264 dated stones, rho 0.51, with latitude held constant 0.51. Half B: 236 dated stones, rho 0.54, with latitude held constant 0.53. Both Holm-corrected p below 10^{-17}: the halves *hold*. The style groups on average along the axis: RAK -0.79, Fp -0.89, Pr1 -0.46, Pr2 -0.18, Pr3 -0.01, Pr4 0.51, Pr5 0.80: the sequence is that of the stylistic chronology, except that RAK and Fp change places. In plain words: if one orders the stones only by what is written and how, they fall in roughly the same order as when one orders them by the ornament; the association is moderately strong and equally strong in both halves.

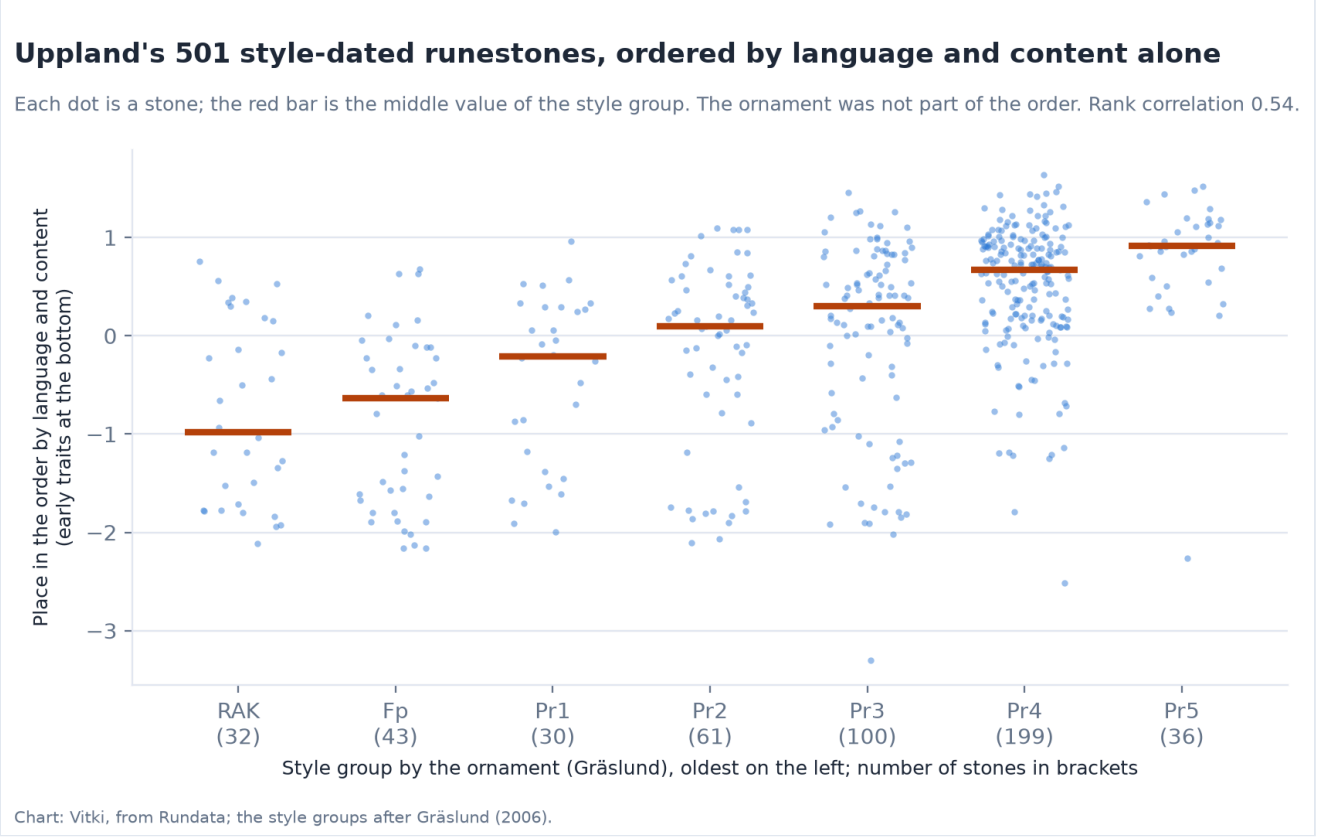


Figure 1. Uppland's 501 style-dated runestones, ordered by language and content alone, against the style group. Each dot is a stone and the bar the middle value of the group: the order rises with the style groups, but the spread within each group is large. Chart: Vitki, after Rundata (ODbL); the style groups after Gräslund (2006).

5.2 Södermanland (replication). 151 dated stones: rho 0.32, Holm-corrected p = $6.3 \cdot 10^{-5}$; *replicated* by the plan's rule. With latitude held constant 0.29, below the plan's limit of 0.3, so the association may be partly geographical. Within hundreds 0.24 (11 hundreds).

5.3 The sensitivity analyses step by step. The project's fixed series of sensitivity analyses is built for associations between two traits and has not been run on the seriation. Step by step, with what

comes closest:

- The halves: tested (5.1); they hold.
- Adjusted for province: not tested; the seriation is made within one province, and Södermanland is tested on its own (5.2).
- The carver as a random effect: not tested. Closest is the correlation within carvers: only 0.25 (13 carvers with at least five dated stones, 251 stones; exploratory). Of the spread in dating among these stones only 18 percent lies within the carvers (computed on the stones' rank order in time); the rest separates the carvers, so within one carver there is little time to measure.
- The parish as a random effect: not tested. Closest is the correlation within hundreds: 0.51 (23 hundreds; exploratory).
- Location as a spatial surface: not tested. Closest is the partial correlation with latitude held constant: 0.54.
- Without one province at a time: not applicable, since the seriation concerns one province; instead the replication in Södermanland.
- Without moved stones: not tested.
- Model check: not tested; the correspondence analysis has no corresponding check of residuals.
- Adjustment together with all 48 associations the project reports: done, with the seriation's own p since it has no sensitivity analyses of that kind; $q < 0.001$. The Holm correction covers the plan's three tests.
- The variants of the traits (exploratory): without cross 0.55, without journey abroad 0.54, without the traits that change over time 0.54, without all of these 0.53.

5.4 Dating individual stones. If the axis is used to estimate years, the mean error is 21 years when each stone is estimated without itself, against 25 years for guessing the mean year: the seriation orders the stones, but it does not date a single stone much better than a guess.

6. Discussion

What the finding says. Language, formulas and content changed in the same sequence as the ornament in Uppland, and it holds equally strongly in two halves of the material. Since the ornament is not included, it is support for Gräslund's relative chronology that does not rest on the ornament; how much of it runs through the carvers the material cannot tell (5.3). It also holds without the crosses, which are part of the picture, without the trait journey abroad (which the Ingvar and England stones carry and which could otherwise anchor the time) and without the spelling traits whose change is measured separately.

Against earlier research. The stylistic chronology is the basis on which most datings of Swedish runestones rest (Gräslund 2006). The result says that it is not only an ordering of ornament: language and content follow the same time. In Södermanland, where the chronology is applied but was not built, the same sequence is found but a little over half as strong.

Weaknesses, in order of weight.

1. **The sequence is weak within carvers.** Within one and the same carver the correlation is only 0.25. Ornament and language both follow the carver, and the carvers worked at different times; part of the agreement may therefore be that the seriation orders carvers. But each carver's stones span a short time (18 percent of the spread in dating lies within the carvers, and 2 of the 13 carvers have stones in a single style group), so a weak correlation within carvers is expected even if language follows time. Within hundreds it holds (0.51), so it is not an ordering of districts.
2. **Not pre-registered.** The seriation in Uppland was known when the plan for the halves and Södermanland was locked, and Uppland was chosen after the whole material had shown a regional first axis. The halves and Södermanland are therefore the actual test.
3. **Södermanland may be geography.** With latitude held constant the correlation is 0.29, below the plan's limit.
4. **Not a dating.** The mean error for years is only somewhat better than guessing the mean year.

What would overturn the finding. If the correlation disappeared within carvers also in a larger material with more carvers who worked long, or if a dating independent of both ornament and language gave another sequence.

7. Conclusion

Ordered by language, formulas and content, the runestones of Uppland fall in the same sequence as in the stylistic chronology, rho 0.54, and it holds in both halves. That is support for the relative chronology that does not rest on the ornament; how much of it runs through the carvers the material cannot tell, and in Södermanland the association is weaker and partly geographical. The next step is to test the seriation within carvers who worked over a long time, and in Östergötland and Västergötland when the dating there becomes more secure.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se). Code state: commit 4f80a63 in https://github.com/Precatio/runes (Vitki 3.0.0 plus later changes); the report's builder for the versions of 2026-11-05 from commit 6eae0af. The code, the analysis plan and the result files are there under GPL-3.0.

- The seriation: `r/chronology.R` (results `chronology.json` under the material's version `e83202e01a5e6d27`).
- The plan for the halves and Södermanland: `analysplaner/2026-10-31-seriation-halvor-sodermanland.md`; the code `utdata/seriation/replikering.R`, the results `replikering.json` and `replikering_resultat.md`.
- The report is built by `utdata/blogg/build/build_rapport_seriation.py`, which reads every number from the result files and stops if anything disagrees.

- In Vitki the seriation is shown under Statistics, Dating. Glossary of the statistical terms: https://vitkirunor.se/en/forskningsrapporter/begrepp.

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Change log

- **Version 2, 2026-11-01.** Figure 1 added: every style-dated stone in Uppland, ordered by language and content, against the style group. No numbers, interpretations or conclusions have changed.
- **Version 3, 2026-11-05.** Corrections after the review before archiving in Zenodo. The variant without journeys abroad drops the trait journey abroad, not the Ingvar and England stones, as the text said. That the seriation's first axis in the whole Swedish material follows regional differences and time only weakly, the reason why it is made on Uppland, was left out. ”Independent support” and ”two independent clocks” said more than the material bears, since the correlation within carvers is only 0.25. Södermanland's association is a little over, not about, half as strong. The sensitivity analyses not tested are now listed step by step (5.3, which has changed places with Södermanland as the template prescribes), and earlier research on single traits against the style groups is mentioned. The chart of the rank correlations showed them as percentages and is removed; the numbers are in the text. Code state, data versions, absolute links and a source in the caption were added. No analysis was redone and no earlier figure changed.
- **Version 4, 2026-11-05.** Corrections after a second review before archiving in Zenodo. The adjustment for all 48 associations the project reports was said not to be tested; it is done, with the seriation's own p, and gives $q < 0.001$ (abstract and 5.3). ”Largely through the carvers” said more than the material bears: each carver's stones span a short time (only 18 percent of the spread in dating lies within the carvers, and 2 of 13 carvers have stones in a single style group), so a weak correlation within carvers is expected even if language follows time; this now stands in the abstract, the discussion, weakness 1, the conclusion and the essentials box. Version 3 added, without saying so, that ordering many traits at once is new; it now says that we have found no such

study. The English sentence on why the seriation is made on Uppland was ambiguous about the carvers. The code state now also names the commit holding the report's builder. No analysis was redone; the share of 18 percent is computed from the seriation's own results, and no earlier figure changed.

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Cite as

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